sectors. The method is most commonly used for estimating the impacts of positive or negative economic shocks and analyzing the ripple effects throughout an economy.

The foundation of Input–Output analysis involves input–output tables. Such tables include a series of rows and columns of data that quantify the supply chain for sectors of the economy. Industries are listed in the heads of each row and each column. The data in each column corresponds to the level of inputs used in that industry's production function. For example the column for auto manufacturing shows the resources required for building automobiles (ie., requirement of steel, aluminum, plastic, electronic etc.,). Input–Output models typically includes separate tables showing the amount of labour required per rupee unit of investment or production.

Consider a simple economic model consisting of two industries A_1 and A_2 where each produces only one type of product. Assume that each industry consumes part of its own output and rest from the other industry for its operation. The industries are thus interdependent. Further assume that whatever is produced that is consumed. That is the total output of each industry must be such as to meet its own demand, the demand of the other industry and the external demand (final demand).

Our aim is to determine the output levels of each of the two industries in order to meet a change in final demand, based on knowledge of the current outputs of the two industries, of course under the assumption that the structure of the economy does not change.

Let a_{ij} be the rupee value of the output of A_i consumed by A_j, $i, j = 1, 2$

Let x_1 and x_2 be the rupee value of the current outputs of A_1 and A_2 respectively.

Let d_1 and d_2 be the rupee value of the final demands for the outputs of A_1 and A_2 respectively.

The assumptions lead us to frame the two equations

$$a_{11} + a_{12} + d_1 = x_1;$$
$$a_{21} + a_{22} + d_2 = x_2 \qquad \ldots (1)$$

Let $b_{ij} = \frac{a_{ij}}{x_j}$ $i, j = 1, 2$

$$b_{11} = \frac{a_{11}}{x_1}; \quad b_{12} = \frac{a_{12}}{x_2};$$

$$b_{21} = \frac{a_{21}}{x_1}; \quad b_{22} = \frac{a_{22}}{x_2}$$

The equations (1) take the form

$$b_{11}x_1 + b_{12}x_2 + d_1 = x_1$$

$$b_{21}x_1 + b_{22}x_2 + d_2 = x_2$$

The above equations can be rearranged as

$$(1 - b_{11})x_1 - b_{12}x_2 = d_1$$

$$-b_{21}x_1 + (1 - b_{22})x_2 = d_2$$

The matrix form of the above equations is

$$\begin{bmatrix} 1 - b_{11} & -b_{12} \\ -b_{21} & 1 - b_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$$\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$$(1-B)X = D$$

where $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$, $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

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